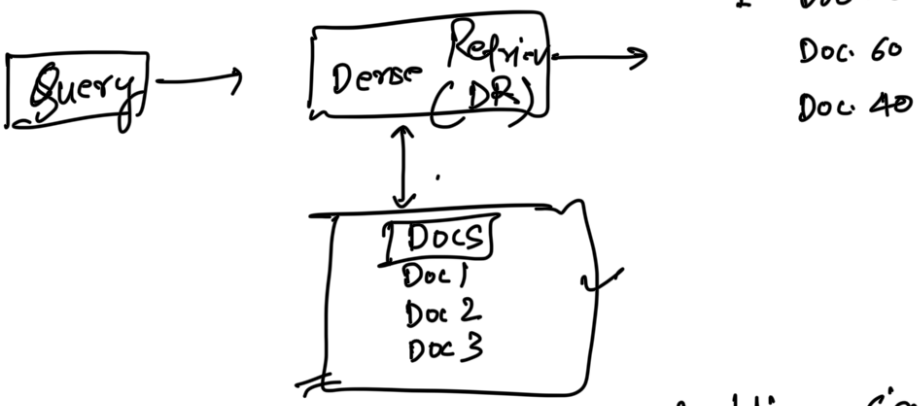
15/3/26

LoRA / QLoRA / Flash Attention

RAG: Retrieval Augmented Generation

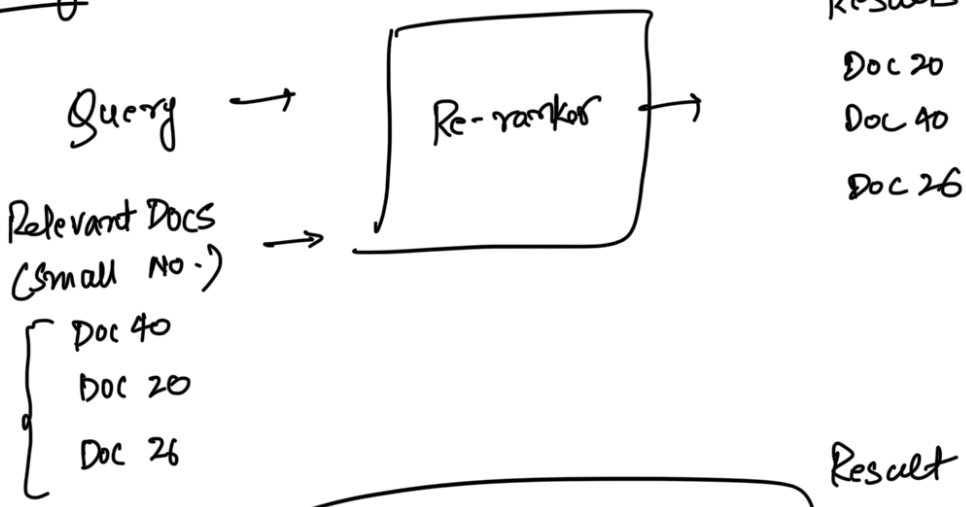
Semantic Search

Dense Retrieval (DR)

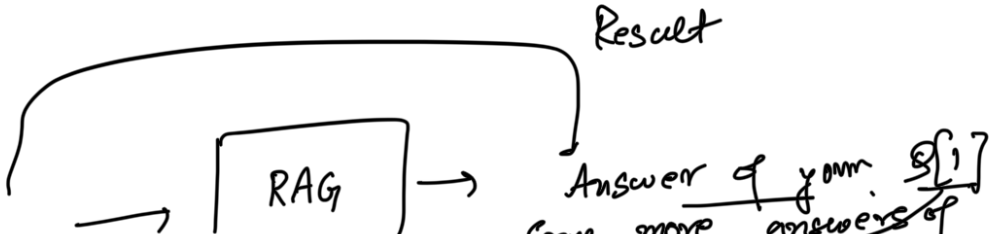


It is relying on text-embedding similarity to retrieve results.

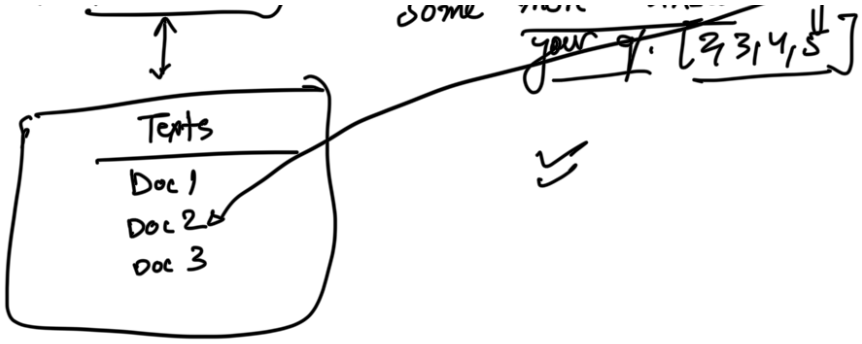
Reranking :



RAG

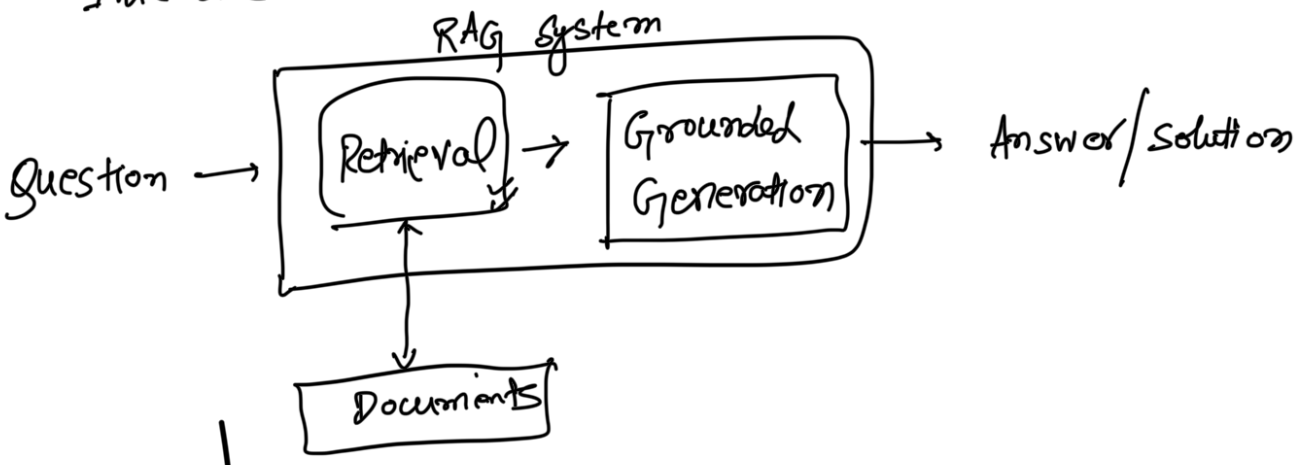


question

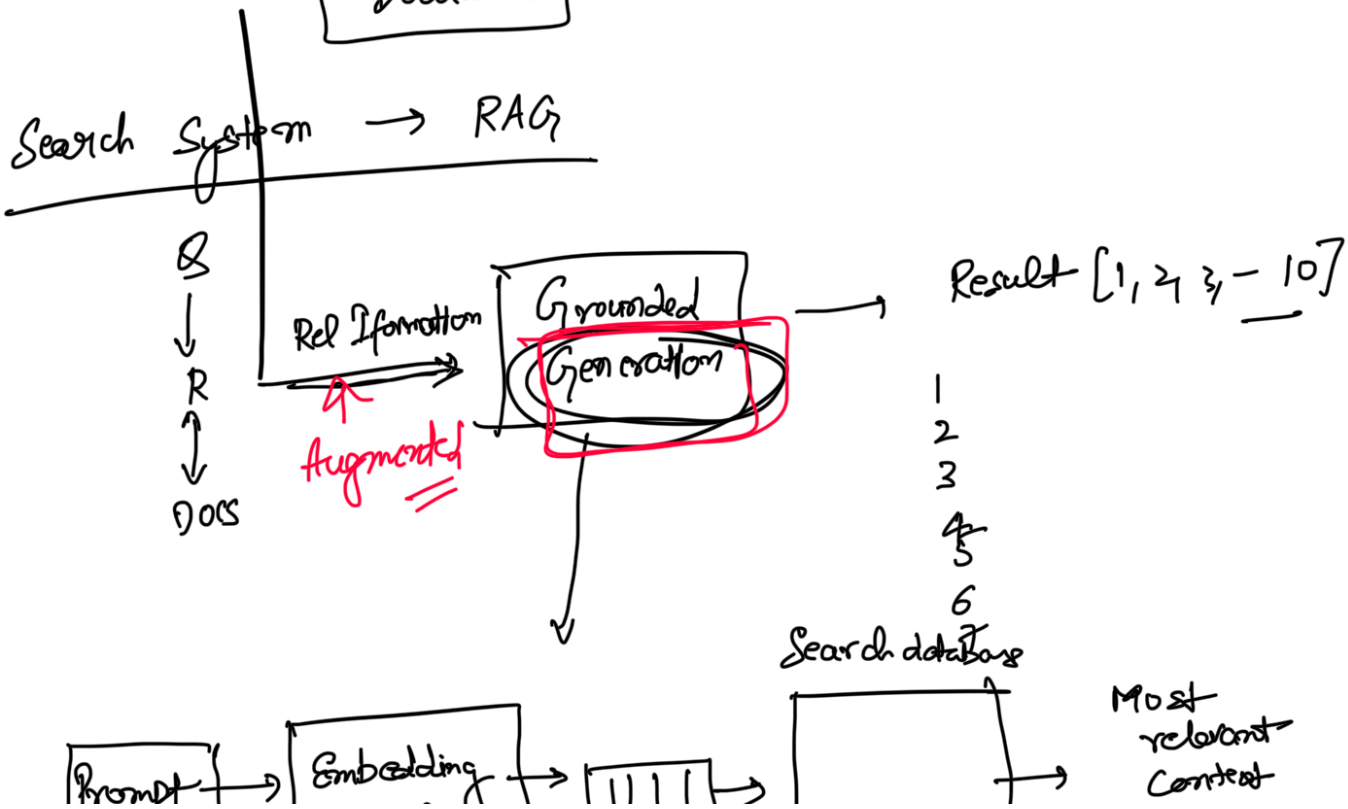


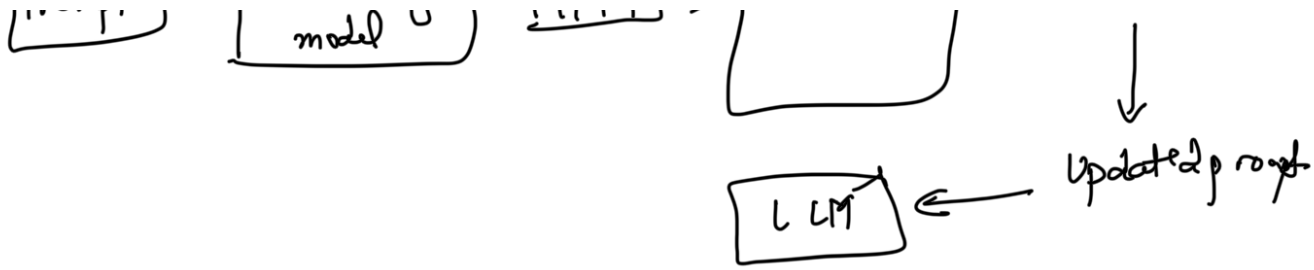
Reduce hallucinations ✓
Increase factuality [Refs are checked]
Application specific data set ✓

66 Retrieval Augmented Generation for Knowledge-Intensive NLP Tasks" (2020)



Search System → RAG





Query Rewriting:

"you are a Sentiment analysis engine. ~~I want you to analyse the~~ mail received ~~to return the~~ sentiment of the user. ~~You can name it as~~ happy, neutral and sad"

Analyse the sentiment of ~~Email~~ as happy, neutral and sad

Multi-query RAG:

Compare the performance of X in 2025 vs 2026

performance of X in 2025
performance of X in 2026

Multi-hop RAG:

Q1? Q2?
Ans 1 → Answer 2

Who ~~are~~ were the best ~~phys~~ teachers in 2021? Do they teach physics?

→ X1
X2
X3

Query routing: helping the model to search multiple datasets.

RAG Evaluations:

"Evaluating verifiability in generative search engines" 2023

1. Fluency → ✓
2. Perceived utility → helpful / informative ✓
3. Citation Recall → citations ref to ✓
4. Citation precision → No. of citations ✓

(LLM as a judge)

Faithfulness → Answer is consistent ✓

Answer relevance → relevant to the question